How the project uses D-Wave

Code walk-through, modes of use, checks, estimates, budget guard, and the 1-second test

`Projects_master_HPQC/scripts/QuantumComputing/Dimod`

Abstract

This document explains, file by file and line by line, how the SK-game campaign talks to D-Wave:

- which Ocean functions it uses, what they can do, and how the code harnesses them;
- the six ways ("modes") of running an annealer in this project;
- what `qpu --check` actually does, and how the QPU time is estimated (three levels, with the formulas);
- how the budget guard (`--budget-seconds`) makes sure a run never exceeds a given QPU time;
- how the results are turned into success probabilities, time-to-solution and the Zephyr/Pegasus comparison.

The code excerpts are read directly from the source files when the document is compiled, so the line numbers are those of the version in the repository. Section 16 gives the exact commands for a test that uses at most one second of QPU time, followed by a plan to spend the 180 s of this month before 4 November.

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1 Where the D-Wave code lives

All paths are relative to `Dimod/src/unlearnable_games_explanation/`.

file	role
`skgame/optim/problem.py`	the SK problem W; `to_bqm()` builds the dimod model
`skgame/optim/embedding.py`	topologies (Zephyr, Pegasus), embedding strategies (clique, heuristic), chain statistics, embedding cache
`skgame/optim/backends.py`	`AnnealBackend`: one object per `device:topology:strategy`; builds the sampler, embeds, samples, collects statistics, estimates time
`skgame/optim/quantum.py`	`QuantumAnnealingSolver`: the older single-solver path (`DWaveCliqueSampler`/`EmbeddingComposite`), used by the planner benchmark
`skgame/optim/solvers.py`	`DimodSamplerSolver` (generic dimod wrapper) and the classical dimod samplers
`main_simulation.py`	the campaign: stages `init`, `instances`, `classical`, `embed`, `qpu`/`sim`, `analyze`; the QPU budget guard; the analysis
`tests/test_embedding.py`, `tests/test_campaign.py`	tests of all the above, including the QPU path against D-Wave's mock QPU

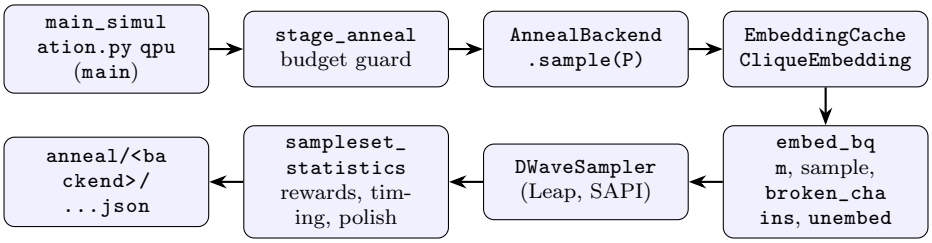


Figure 1: The path of one QPU call through the code.

2 The Ocean SDK: what each package can do, and what we use it for

D-Wave's Python SDK (*Ocean*) is a set of packages, all installed by `cineca/requirements.txt`. For each package: what it offers, then exactly how this project uses it.

2.1 `dwave-cloud-client`: connection, solvers, configuration, timing model

Capabilities.

- it talks to the Leap *Solver API* (SAPI) over HTTPS;
- it reads the token from the configuration file or from the environment (`DWAVE_API_TOKEN`);
- it selects solvers with *feature filters* such as `solver={"topology__type": "zephyr"}`;
- it exposes every solver's `properties`, including the timing data;
- it estimates the cost of a problem (`estimate_qpu_access_time`);
- it submits problems asynchronously;
- it provides the `dwave` command line: `config`, `auth`, `leap`, `solvers`, `ping`, `setup`.

How we use it.

- Through `DWaveSampler` (next subsection), which wraps a cloud-client `Solver`.

- `solver.name` identifies the QPU in the logs and keys the embedding cache, e.g. `Advantage2_system1`.
- `solver.estimate_qpu_access_time` is the second level of our time estimate (§6).
- On the command line: `dwave config create`, `dwave setup --oob`, `dwave solvers -la`, `dwave ping` (`Dimod/LEAP_TOKEN.md`).

2.2 `dwave-system`: samplers for the QPU, composites, mock

Capabilities.

- `DWaveSampler`: a dimod sampler bound to one structured QPU. It exposes `nodelist`/`edgelist` (the working qubits and couplers), `properties`, `parameters` (the accepted keyword arguments) and `to_networkx_graph()`. Its `sample()` accepts `num_reads`, `annealing_time`, `anneal_schedule`, `auto_scale`, `answer_mode`, `h_gain_schedule`, `anneal_offsets`, `programming_thermalization`, `readout_thermalization`, ...
- Composites that turn a structured sampler into one that accepts any problem:
  - `EmbeddingComposite` finds a heuristic embedding on every call;
  - `FixedEmbeddingComposite` uses a given embedding;
  - `LazyFixedEmbeddingComposite` finds one embedding and reuses it;
  - `DWaveCliqueSampler` uses cached all-to-all clique embeddings, scales with the extended J range and computes chain strengths.
- `dwave.system.testing.MockDWaveSampler`: an offline imitation of a QPU of any topology and size, which returns realistic timing fields.

How we use it.

- `DWaveSampler(solver={"topology__type": ...})` picks a Zephyr or Pegasus QPU (code below).
- `to_networkx_graph()` gives the *real* working graph. The embedding is computed on it, so it avoids dead qubits.
- Every QPU call does explicitly what `FixedEmbeddingComposite` does: `embed_bqm`, then `DWaveSampler.sample`, then `unembed_sampleset`, with our cached clique embedding. Doing it by hand keeps the *physical* reads, from which the broken chains of every read are counted (§8). The code was checked to give read-for-read the same results as the composite. The Zephyr and Pegasus back-ends go through identical code and differ only in hardware and embedding.
- `DWaveCliqueSampler` and `EmbeddingComposite` remain available in `QuantumAnnealingSolver` (§3).
- `MockDWaveSampler` is used in the tests and in the dry runs.

`skgame/optim/backends.py`, lines 88–118: building the child sampler (QPU, embedded simulator or logical simulator) and reading its hardware graph

```
88      # ------------------------------------------------------------- hardware graph ----
89      @property
90      def child(self):
91          """The structured sampler the embedding maps onto."""
92          if self._child is None:
93              if self.device == "qpu":
94                  from dwave.system import DWaveSampler
95                  s = self.settings
96                  self._child = DWaveSampler(solver=s.solver_name or self.topology.solver_filter)
97              elif self.embedded:
```

```
                import dimod
                from dwave.samplers import SimulatedAnnealingSampler
                G = self._target = self.topology.ideal_graph()
                self._child = dimod.StructureComposite(SimulatedAnnealingSampler(), list(G.nodes), list(G.edges)
    )
            else:
                from dwave.samplers import SimulatedAnnealingSampler
                self._child = SimulatedAnnealingSampler()
        return self._child

    @property
    def target(self):
        """networkx graph of the working qubits and couplers."""
        if self._target is None and self.embedded:
            child = self.child                # building a sim child also sets the ideal graph
            if self._target is not None:
                pass
            elif hasattr(child, "to_networkx_graph"):
                self._target = child.to_networkx_graph()   # real working graph, keeps the topology metadata
            else:
                raise ValueError(f"{self.spec}: cannot read the hardware graph of {type(child).__name__}")
        return self._target
```

2.3 `dimod`: the problem and the results

Capabilities.

- `BinaryQuadraticModel` in SPIN or BINARY form;
- `SampleSet`, which holds the reads: `record.sample`, `record.energy`, `record.num_occurrences`, extra vectors such as `chain_break_fraction`, plus `info` (timing, embedding context). It also offers `aggregate()`, `first` and `resolve()`;
- a common `Sampler` interface (`sample(bqm, **kwargs)`, `parameters`);
- helper composites such as `StructureComposite`, and reference samplers such as `ExactSolver`.

How we use it.

- The SK problem becomes an Ising BQM whose energy is exactly $H(S) = -\sum_{i<j} W_{ij} S_i S_j$.
- `StructureComposite` restricts simulated annealing to the Zephyr or Pegasus graph (the `sim:*:clique` back-ends).
- `ExactSolver` checks the statistics code in the tests.

`skgame/optim/problem.py`, lines 45–50: the SK problem as a dimod BQM ($h = 0$, $J_{ij} = -W_{ij}$)

```
    def to_bqm(self):
        """dimod BinaryQuadraticModel (SPIN) with E = H(S). Lazy import."""
        import dimod
        iu = np.triu_indices(self.N, 1)
        quad = {(int(i), int(j)): -float(self.W[i, j]) for i, j in zip(*iu)}
        return dimod.BinaryQuadraticModel({}, quad, 0.0, dimod.SPIN)
```

2.4 `minorminer` and `dwave-graphs`: embeddings and hardware graphs

Capabilities.

- `minorminer.busclique.find_clique_embedding` returns all-to-all embeddings made of line segments, optimised for chain length and avoiding defects, with an on-disk cache;
- `busgraph_cache(G).largest_clique()` gives the largest K_N the graph supports;
- `minorminer.find_embedding` is the general heuristic;

- `dwave.graphs` (formerly `dwave_networkx`) provides `zephyr_graph`, `pegasus_graph`, coordinate converters and drawing functions.

How we use it.

- The clique strategy and its maximum size (below), which give the `largest_problem` printed by `qpu --check`.
- The heuristic baseline.
- The ideal graphs for the simulator and for the figures in `visualization/embedding_explanation`.

`skgame/optim/embedding.py`, lines 101–115: the all-to-all strategy: embedding and largest clique

```
101 class CliqueEmbedding(EmbeddingStrategy):
102     """Native all-to-all embedding (minorminer.busclique): every chain has the same, minimal-ish
103     length (Zephyr: about N/16 + 2 qubits; Pegasus: about N/12 + 2), deterministic and fast.
104     This is what DWaveCliqueSampler uses internally."""
105
106     name = "clique"
107
108     def embed(self, N, target):
109         from minorminer.busclique import find_clique_embedding
110         emb = find_clique_embedding(N, target, use_cache=True)
111         return {int(k): [int(q) for q in v] for k, v in emb.items()}
112
113     def max_size(self, target):
114         from minorminer.busclique import busgraph_cache
115         return len(busgraph_cache(target).largest_clique())
```

`skgame/optim/embedding.py`, lines 157–164: chain statistics; `n_qubits` feeds the readout-time model

```
157 def chain_stats(embedding: dict) -> dict:
158     """Chain lengths of an embedding: the main predictor of chain breaks and of the energy
159     compression imposed by the chain couplings."""
160     if not embedding:
161         return {"n_qubits": 0, "chain_len_mean": float("nan"), "chain_len_max": 0, "chain_len_min": 0}
162     L = np.array([len(c) for c in embedding.values()])
163     return {"n_qubits": int(L.sum()), "chain_len_mean": float(L.mean()),
164             "chain_len_max": int(L.max()), "chain_len_min": int(L.min())}
```

2.5 `dwave.embedding` (inside `dwave-system`): chains in and out

Capabilities.

- `embed_bqm` maps a logical BQM onto qubits: it splits each J_{ij} over the couplers between two chains and adds $-c$ on the chain couplers;
- `unembed_sampleset` maps reads back to logical spins;
- `chain_strength.uniform_torque_compensation` gives $c = p\, J_{\mathrm{rms}} \sqrt{\bar{d}}$;
- `chain_breaks`: `majority_vote`, `weighted_random`, `discard`, `MinimizeEnergy`.

How we use it.

- The chain strength is `partial(uniform_torque_compensation, prefactor=p)`, with $p =$ `chain_strength_prefactor`.
- The chain-break method is chosen by name.
- `MinimizeEnergy` has to be wrapped in a named function, because `unembed_sampleset` reads `__name__`. Testing revealed this.

skgame/optim/backends.py, lines 151–156: the keyword arguments for the physical sampler

```
151     def sample_kwargs(self) -> dict:
152         """Keyword arguments for the (physical) sampler itself."""
153         s = self.settings
154         if self.device == "qpu":
155             return {"num_reads": s.num_reads, "annealing_time": s.annealing_time}
156         return {"num_reads": s.sim_num_reads, "num_sweeps": s.sim_num_sweeps}
```

skgame/optim/backends.py, lines 300–310: chain-break resolution by name

```
300 def _chain_break_method(name: str, bqm, emb):
301     from dwave.embedding import chain_breaks as cb
302     if name == "minimize_energy":
303         resolver = cb.MinimizeEnergy(bqm, emb)
304 
305         def minimize_energy(samples, chains):    # unembed_sampleset needs a __name__
306             return resolver(samples, chains)
307         return minimize_energy
308     if name in ("majority_vote", "weighted_random", "discard"):
309         return getattr(cb, name)
310     raise ValueError(f"chain_break_method must be one of {CHAIN_BREAK_METHODS}")
```

2.6 `dwave-samplers` and `dwave-preprocessing`

Capabilities.

- classical samplers with the dimod interface: `SimulatedAnnealingSampler`, `TabuSampler`, `SteepestDescentSolver`, `TreeDecompositionSolver`, `RandomSampler`;
- `SpinReversalTransformComposite` (gauge transforms) and other pre-processing.

How we use it.

- SA, tabu and steepest descent are classical baselines (`solvers.py`).
- SA is also the engine of the `sim:logical` and `sim:*:clique` back-ends.
- Spin-reversal transforms are discussed as a mitigation (`explanation_project.pdf`, noise section) but are not yet wired in.

3 The modes of use

mode	what runs	when to use it	QPU time / token
`sim:logical`	SA on K_N directly (no hardware graph)	reference for the annealing pipeline	none / no
`sim:zephyr:clique`, `sim:pegasus:clique`	the *embedded* problem, chains included, solved by SA on the ideal graph (`StructureComposite`)	test the whole embedding path; classical view of chain effects	none / no
mock QPU (tests)	`MockDWaveSampler` in place of `DWaveSampler`, same code path as the QPU	tests, dry runs	none / no
`qpu:zephyr:clique`, `qpu:pegasus:clique`	real QPU: `embed_bqm` + `DWaveSampler` + `unembed_sampleset` with our clique embedding	the campaign	yes / yes
`qpu:*:minorminer`	real QPU with a heuristic embedding	baseline only	yes / yes

mode	what runs	when to use it	QPU time / token
`QuantumAnnealingSolver`	`DWaveCliqueSampler` (clique, extended J range) or `EmbeddingComposite` (heuristic each call)	the planner benchmark (`skgame/optim/benchmark.py`)	yes / yes

`skgame/optim/quantum.py`, lines 49–58: the single-solver path: `DWaveCliqueSampler` or `EmbeddingComposite`

```
    def make_sampler(self):
        if self.use_simulator:
            from dwave.samplers import SimulatedAnnealingSampler
            return SimulatedAnnealingSampler()
        if self.clique:
            from dwave.system import DWaveCliqueSampler
            return DWaveCliqueSampler(solver=self.solver_name) if self.solver_name else DWaveCliqueSampler()
        from dwave.system import DWaveSampler, EmbeddingComposite
        base = DWaveSampler(solver=self.solver_name) if self.solver_name else DWaveSampler()
        return EmbeddingComposite(base)
```

The campaign embeds explicitly (the steps of `FixedEmbeddingComposite`) rather than using `DWaveCliqueSampler`, so that both topologies run *identical* code: same embedding strategy, same chain-strength rule, same scaling. Their results then differ only because of the hardware.

4 Life of one QPU call, line by line

`skgame/optim/backends.py`, lines 158–243: `AnnealBackend.sample`: embed, sample, keep the physical reads, unembed

```
    def sample(self, P: PlannerProblem, rng=None, label: str | None = None) -> dict:
        """One call. For embedded back-ends this does explicitly what FixedEmbeddingComposite does
        (embed -> sample -> unembed), so that the PHYSICAL reads are available in between: from
        them `broken_chains` tells, read by read, which logical spins had a broken chain."""
        rng = np.random.default_rng(rng)
        bqm = P.to_bqm()
        kw = self.sample_kwargs()
        params = getattr(self.child, "parameters", {}) or {}
        if self.device == "sim" and "seed" in params:
            kw["seed"] = int(rng.integers(2**31))
        if label and "label" in params:                    # shown in the Leap dashboard next to the problem id
            kw["label"] = label[:256]
        broken, phys_counts, cs_used, emb, target = None, None, None, None, None
        t0 = time.perf_counter()
        if self.embedded:
            import dimod
            from dwave.embedding import EmbeddedStructure, broken_chains, unembed_sampleset
            from dwave.embedding.chain_strength import uniform_torque_compensation
            emb = self.embedding(P.N)
            structure = EmbeddedStructure(self.child.edgelist, emb)
            target = structure.embed_bqm(bqm, chain_strength=partial(
                uniform_torque_compensation, prefactor=self.settings.chain_strength_prefactor),
                smear_vartype=dimod.SPIN)
            cs_used = structure.chain_strength
            to_send, scaling = self._scale(target, emb, float(cs_used))
            if scaling["mode"] != "auto_scale":
                kw["auto_scale"] = False
            if self.flux_biases and "flux_biases" in params:
                nq = int((getattr(self.child, "properties", {}) or {}).get("num_qubits", 0)) or \
                    (max(map(int, self.child.nodelist)) + 1)
                fb = [0.0] * nq
                for q, v in self.flux_biases.items():
                    fb[int(q)] = float(v)
                kw["flux_biases"] = fb
                if "flux_drift_compensation" in params:
                    kw["flux_drift_compensation"] = True
            phys = self.child.sample(to_send, **kw)
            phys.resolve()
            chains = [emb[i] for i in range(P.N)]
            broken = np.asarray(broken_chains(phys, chains), bool)          # (physical rows, N)
            phys_counts = np.asarray(phys.record.num_occurrences, int)
```

```
            ss = unembed_sampleset(phys, structure, bqm,
                                   chain_break_method=_chain_break_method(self.settings.chain_break_method, bqm,
 emb),
                                   chain_break_fraction=True)
            ss.info.update({k: v for k, v in phys.info.items() if k not in ss.info})
        else:
            ss = self.child.sample(bqm, **kw)
            ss.resolve()
        wall = time.perf_counter() - t0
        out = sampleset_statistics(ss, P, wall, polish=self.settings.polish, broken=broken, phys_counts=
phys_counts,
                                   discard=self.settings.chain_break_method == "discard")
        info = out["info"]
        info.update(backend=self.spec, solver=self.solver_id(), settings=asdict(self.settings),
                    sample_kwargs={k: v for k, v in kw.items() if k not in ("seed", "flux_biases")})
        if "flux_biases" in kw:
            v = np.array([x for x in kw["flux_biases"] if x != 0.0])
            info["flux_biases"] = {"source": self.flux_biases_source, "n_nonzero": int(v.size),
                                   "rms": float(np.sqrt(np.mean(v ** 2))) if v.size else 0.0}
        if self.record_qubit_magnetization and self.embedded:
            w = np.asarray(phys.record.num_occurrences, float)
            m = (np.asarray(phys.record.sample, float) * w[:, None]).sum(0) / w.sum()
            info["qubit_magnetization"] = {str(int(q)): float(x) for q, x in zip(phys.variables, m)}
        for k in ("problem_id", "problem_label"):
            if k in ss.info:
                info[k] = str(ss.info[k])
        info["solver_properties"] = self.solver_snapshot()
        if emb is not None:
            info["embedding"] = chain_stats(emb)
            info["embedding_used"] = {str(i): [int(q) for q in emb[i]] for i in range(P.N)}
            info["chain_lengths"] = [len(emb[i]) for i in range(P.N)]
            info["chain_strength"] = float(cs_used) if np.isscalar(cs_used) else None
            info["max_abs_logical_coupling"] = float(np.max(np.abs(P.W)))
            q = np.array(list(target.quadratic.values()), float)
            lo, hi = (info["solver_properties"].get("j_range") or [-1.0, 1.0])
            # what auto_scale does here (no linear terms): divide all couplers by the factor that
            # brings the most extreme one to the end of the allowed range [lo, hi]
            worst = max(q.max(initial=0.0) / hi, q.min(initial=0.0) / lo)
            uv = np.array(list(target.quadratic.keys()), dtype=np.int64)
            out["_physical"] = {"u": uv[:, 0], "v": uv[:, 1], "J": q,
                                "h_q": np.array(list(target.linear.keys()), dtype=np.int64),
                                "h": np.array(list(target.linear.values()), float), "offset": float(target.
offset)}
            info["programmed"] = {"max_abs_J_before_scaling": float(np.abs(q).max()),
                                  "auto_scale_divisor": float(worst) if scaling["mode"] == "auto_scale"
                                  else float(scaling["divisor"]), "scaling": scaling,
                                  "n_qubits": int(target.num_variables), "n_couplers": int(target.
num_interactions)}
        return out
```

What happens, in order:

1. `P.to_bqm()`: the SK problem as an Ising BQM.
2. `self.embedding(P.N)`: the clique embedding of K_N on the solver's real graph, read from `embeddings/<solver>/clique/N....json` or computed and cached.
3. `EmbeddedStructure(child.edgelist, emb).embed_bqm(bqm, chain_strength=...)` computes c with the chain-strength callable, splits each J_{ij} over the couplers between two chains and puts $-c$ on the chain couplers.
4. `DWaveSampler.sample(target, num_reads, annealing_time)` submits the physical problem to SAPI. `auto_scale` is on by default and divides by the largest coefficient.
5. `broken_chains(phys, chains)`: for every physical read and every logical spin, whether its chain is broken (qubits disagreeing).
6. `unembed_sampleset(phys, structure, bqm, chain_break_method=..., chain_break_fraction=True)` maps the reads back to logical spins.
7. `ss.resolve()` waits for the asynchronous answer.

8. `sampleset_statistics` recomputes every reward from the spins: it is exact whatever scaling was applied. It aggregates identical reads, records `info["timing"]` (in μs) and the read-averaged chain-break fraction, and optionally polishes every distinct read with greedy descent.

9. The embedding statistics, the chain strength actually used (from `embedding_context`) and $\max|W_{ij}|$ are added.

`skgame/optim/backends.py`, lines 313–385: statistics of a sample set, including the broken chains of every read

```
313 def sampleset_statistics(ss, P: PlannerProblem, wall: float, polish: bool = False, broken=None,
314                         phys_counts=None, discard: bool = False) -> dict:
315     """Distinct rewards and their read counts (best first), best spins, QPU timing, chain breaks.
316
317     Rewards are recomputed from the spins, so they are exact in units of R_N whatever scaling
318     the sampler applied. With ‘polish‘, every distinct read is also relaxed by greedy single-flip
319     ascent (a classical post-processing step that repairs the residual errors of chain breaks
320     and of analogue noise) and reported separately.
321
322     With ‘broken‘ (physical rows x N booleans from dwave.embedding.broken_chains) the result also
323     has, read by read, how many chains broke:
324       out["reads"]                 distinct (configuration, number of broken chains) pairs:
325                                    reward, polished_reward, n_broken, count
326       info["broken_chains_hist"]   {number of broken chains: number of reads}
327       info["chain_break_count"]    for each logical spin i, in how many reads its chain broke
328       info["discarded_reads"]      reads dropped by chain_break_method="discard"
329     """
330     agg = ss.aggregate()
331     if len(agg) == 0:                       # chain_break_method="discard" dropped every read
332         S = np.zeros((0, P.N))
333     else:
334         cols = [list(agg.variables).index(i) for i in range(P.N)]
335         S = np.asarray(agg.record.sample, float)[:, cols]
336     counts = np.asarray(agg.record.num_occurrences, int)
337     rewards = np.atleast_1d(P.reward(S)) if len(S) else np.zeros(0)
338     order = np.argsort(-rewards)
339     info = {"wall_seconds": wall, "num_reads": int(counts.sum())}
340     timing = ss.info.get("timing")
341     if timing:
342         info["timing_us"] = {k: float(v) for k, v in timing.items() if np.isscalar(v)}
343     if "chain_break_fraction" in ss.record.dtype.names and len(ss):
344         info["chain_break_fraction_mean"] = float(np.average(ss.record.chain_break_fraction,
345                                                              weights=ss.record.num_occurrences))
346     empty = len(S) == 0
347     out = {"rewards": rewards[order].tolist() if not empty else [], "counts": counts[order].tolist(),
348            "best_reward": float(rewards[order[0]]) if not empty else -np.inf,
349            "best_S": _spin_string(S[order[0]]) if not empty else "", "info": info}
350     polish_of = None
351     if polish and empty:
352         out.update(polished_rewards=[], polished_counts=[], polished_best_reward=-np.inf, polished_best_S="")
353         polish_of = {}
354     elif polish:
355         from .solvers import _greedy
356         Sp = np.array([_greedy(P.W, s.copy())[0] for s in S])
357         rp = np.atleast_1d(P.reward(Sp))
358         op = np.argsort(-rp)
359         out.update(polished_rewards=rp[op].tolist(), polished_counts=counts[op].tolist(),
360                    polished_best_reward=float(rp[op[0]]), polished_best_S=_spin_string(Sp[op[0]]))
361         polish_of = {S[k].tobytes(): float(rp[k]) for k in range(len(S))}
362     if broken is not None:
363         nb_phys = broken.sum(axis=1)
364         info["chain_break_count"] = (broken * phys_counts[:, None]).sum(axis=0).astype(int).tolist()
365         hist = {}
366         for nb, c in zip(nb_phys, phys_counts):
367             hist[int(nb)] = hist.get(int(nb), 0) + int(c)
368         info["broken_chains_hist"] = {str(k): v for k, v in sorted(hist.items())}
369         kept = nb_phys == 0 if discard else np.ones(len(nb_phys), bool)
370         info["discarded_reads"] = int(phys_counts[~kept].sum())
371         nb_rows = nb_phys[kept]
372         Sr = (np.asarray(ss.record.sample, float)[:, [list(ss.variables).index(i) for i in range(P.N)]]
373               if len(ss) else np.zeros((0, P.N)))
374         assert len(Sr) == len(nb_rows), "unembedded rows do not match the physical reads"
375         table = {}
376         for s_row, nb, c in zip(Sr, nb_rows, ss.record.num_occurrences):
377             key = (s_row.tobytes(), int(nb))
```

```
            table[key] = table.get(key, 0) + int(c)
        keys = list(table)
        Sk = np.array([np.frombuffer(k[0]) for k in keys]).reshape(len(keys), P.N)
        rk = np.atleast_1d(P.reward(Sk)) if keys else np.zeros(0)
        out["reads"] = {"S": [_spin_string(x) for x in Sk], "reward": rk.tolist(), "n_broken": [k[1] for k in
     keys],
                        "count": [table[k] for k in keys],
                        "polished_reward": [polish_of[k[0]] for k in keys] if polish_of is not None else None}
    return out
```

The `seed` lesson. Samplers differ in which arguments they accept. The QPU rejects unknown ones *server-side*, so a Python `try/except TypeError` would never catch the error. Both wrappers therefore pass `seed` only to samplers that declare it in `parameters`:

`skgame/optim/solvers.py`, lines 379–388: passing `seed` only where it is accepted

```
        # only samplers that declare a 'seed' parameter get one: the QPU rejects unknown kwargs
        # server side (not with a TypeError), so trying and catching is not enough there
        params = getattr(self._sampler, "parameters", None)
        if params is None or "seed" in params:
            kw.setdefault("seed", int(rng.integers(2**31)))
        try:
            ss = self._sampler.sample(bqm, **kw)
        except TypeError:
            kw.pop("seed", None)
            ss = self._sampler.sample(bqm, **kw)
```

5 What `qpu --check` does (and why it costs nothing)

`python main_simulation.py qpu --out DIR --check` runs, for each QPU back-end of `config.json`:

1. `make_backend` builds an `AnnealBackend`.
2. `be.largest_problem()` builds `DWaveSampler(solver={"topology__type": ...})`. This contacts SAPI to choose an online solver of that family and download its properties: **metadata only, no problem is submitted**. The method then reads the working graph and calls `busgraph_cache(G).largest_clique()`.
3. The list of calls is built. N above the largest clique is skipped; already-saved results are skipped (resume).
4. `_estimate_qpu_time` computes, for each N, the clique embedding on the real graph and the number of qubits it uses, then evaluates `solver.estimate_qpu_access_time`: **a local formula, no QPU**.

`main_simulation.py`, lines 466–479: the `--check` estimate

```
def _estimate_qpu_time(be, todo, log) -> None:
    """QPU access time still to spend, from the solver's own timing model (no QPU time used)."""
    if be.device != "qpu":
        log("[anneal] not a QPU back-end: nothing to estimate")
        return
    per_N = {}
    for N in sorted({u[3] for u in todo}):
        per_N[N] = be.estimate_access_time(N)
        if per_N[N] is None:
            log("[anneal] the solver exposes no timing model")
            return
        log(f"[anneal]   N={N:4d}: ~{per_N[N] * 1e3:.0f} ms per call")
    total = sum(per_N[u[3]] for u in todo)
    log(f"[anneal] estimated QPU access time for the {len(todo)} remaining calls: {total:.1f} s")
```

skgame/optim/backends.py, lines 290–297: one call's estimate from the solver's timing model

```
290     def estimate_access_time(self, N: int) -> float | None:
291         """Seconds of QPU access time for one call, from the solver's timing model (no QPU use)."""
292         solver = getattr(self.child, "solver", None)
293         est = getattr(solver, "estimate_qpu_access_time", None)
294         if est is None:
295             return None
296         n_qubits = chain_stats(self.embedding(N))["n_qubits"]
297         return est(n_qubits, num_reads=self.settings.num_reads, annealing_time=self.settings.annealing_time) / 1
     e6
```

back-end	solver	largest N	calls (`full`)	estimated time
`qpu:zephyr:clique`	`Advantage2_system1`	104	260 ($N \leq 104$)	52.3 s
`qpu:pegasus:clique`	`Advantage_system4`	177	400	105.0 s

Your check (27 October). On the real Zephyr chip the largest all-to-all problem is $N = 104$, far below the 184 of the ideal graph. Defects remove intact line segments (`explanation_project.pdf`, embedding step 10).

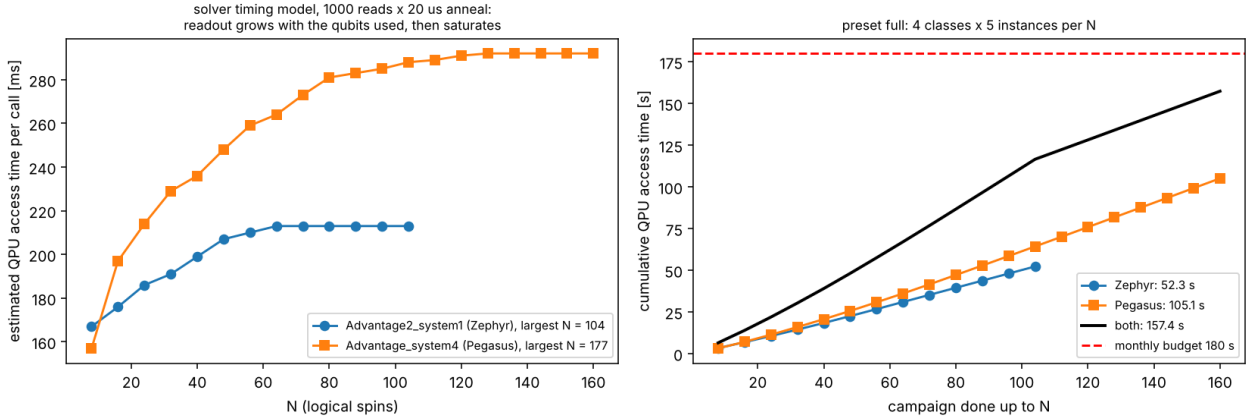


Figure 2: Left: estimated access time per call, from your `--check`. Right: cumulative budget of the `full` campaign: 52.3 s (Zephyr) + 105.0 s (Pegasus) = 157.3 s, against 180 s.

6 How QPU time is estimated: three levels

6.1 Level 1: the rough plan (`main_simulation.py plan`)

For a first idea, before any connection:

$$t_{\text{call}} \approx t_{\text{prog}} + n_{\text{reads}}\,(t_a + t_{\text{ro}} + t_d) = 15\,\text{ms} + 1000\,(20 + 100 + 20)\,\mu\text{s} = 155\,\text{ms}. \tag{1}$$

The constants are generic, and your check shows they **underestimate**: the real per-call times are 157–292 ms. Use level 1 only to compare designs, never to spend.

6.2 Level 2: the solver's timing model (`--check`, budget guard)

D-Wave publishes, in each solver's `properties["problem_timing_data"]`, the parameters of a model that `estimate_qpu_access_time` evaluates. Reading its source (`dwave-cloud-client` 0.14.8) gives

$$t_{\text{access}} = \underbrace{t_{\text{prog}}^{\text{typ}} + t_{\text{prog}}^{\text{therm}}}_{\text{programming}} + n_{\text{reads}}\Big(t_a + t_{\text{ro}}(n_{\text{qubits}}) + t_{\text{delay}} + \max(t_{\text{ro}}^{\text{therm}}, t_{\text{decorr}})\Big), \tag{2}$$

$$t_{\text{ro}}(n_{\text{qubits}}) = \text{piecewise-linear interpolation (often in log–log) of a measured curve.} \tag{3}$$

Two consequences, both visible in your check:

- the cost grows with N, because more qubits mean a longer readout, and then **saturates**;
- Pegasus costs more per call than Zephyr (292 vs 213 ms at large N), because its chains use more qubits.

Above all, the time is **proportional to `num_reads`** apart from the programming term. Halving the reads roughly halves the budget.

6.3 Level 3: the charge (`qpu_access_time`)

After each call the QPU returns its timing. `qpu_access_time` is what Leap charges. It is saved in every result file together with the level-2 estimate (`estimated_access_s`), and `qpu --spent` sums them and prints the ratio actual/estimate:

`main_simulation.py`, lines 448–463: `qpu --spent`: what was actually charged

```
448 def qpu_spent(out: Path, log=print) -> dict:
449     """QPU access time already spent in a campaign, per back-end, from the saved results
450     (the value D-Wave charged for each call), with the ratio to the prediction."""
451     rows = {}
452     for f in sorted((out / "anneal").glob("qpu_*/*/*.json")) if (out / "anneal").exists() else []:
453         info = json.loads(f.read_text())["info"]
454         r = rows.setdefault(f.parent.parent.name, {"calls": 0, "spent_s": 0.0, "est_s": 0.0})
455         r["calls"] += 1
456         r["spent_s"] += info.get("timing_us", {}).get("qpu_access_time", 0.0) / 1e6
457         r["est_s"] += info.get("estimated_access_s", 0.0)
458     total = sum(r["spent_s"] for r in rows.values())
459     for name, r in rows.items():
460         ratio = f"{r['spent_s'] / r['est_s']:.2f}" if r["est_s"] else "n/a"
461         log(f"[spent] {name:22s} {r['calls']:4d} calls  {r['spent_s']:8.3f} s  (actual/estimate = {ratio})")
462     log(f"[spent] total QPU access time in {out}: {total:.3f} s")
463     return rows
```

7 The budget guard: `qpu --budget-seconds S`

`main_simulation.py`, lines 341–372: budget object and per-call estimate

```
341 class QPUBudget:
342     """Hard cap on the QPU access time spent by one command (all back-ends together).
343
344     Before each call the cost is predicted with the solver's own timing model
345     (`estimate_qpu_access_time`) and inflated by a safety factor; the call is made only if
346     spent + inflated estimate <= limit. The factor starts at `safety` and grows if real
347     charges come out larger than predicted (factor >= 1.1 x worst observed actual/estimate).
348     After each call the REAL `qpu_access_time` returned by the QPU is added to `spent`.
349     """
350
351     def __init__(self, limit_s: float | None, safety: float = 1.5):
352         self.limit, self.safety, self.spent, self.worst_ratio = limit_s, safety, 0.0, 0.0
353
354     def factor(self) -> float:
355         return max(self.safety, 1.1 * self.worst_ratio)
356
357     def allows(self, estimate_s: float) -> bool:
358         return self.limit is None or self.spent + self.factor() * estimate_s <= self.limit
359
360     def charge(self, actual_s: float, estimate_s: float) -> None:
361         self.spent += actual_s
362         if estimate_s > 0 and actual_s > 0:
363             self.worst_ratio = max(self.worst_ratio, actual_s / estimate_s)
364
365
366 def _call_estimate(be, N: int, cfg: CampaignConfig) -> float:
367     """Predicted QPU access time of one call [s]. Without a timing model (mock QPU) a
368     deliberately pessimistic formula: 20 ms programming + reads x (anneal + 300 us)."""
369     est = be.estimate_access_time(N) if be.device == "qpu" else 0.0
370     if est is None:
```

```
        est = 0.020 + cfg.num_reads * (cfg.annealing_time * 1e-6 + 300e-6)
    return float(est)
```

`main_simulation.py`, lines 387–445: the anneal stage: guard before every call, real charge after it

```
def stage_anneal(out: Path, cfg: CampaignConfig, spec: str = "sim:logical", keys=None, max_calls=None,
                 solver_name=None, check_only=False, log=print, retries=3, backend=None,
                 budget: QPUBudget | None = None, Ns=None) -> None:
    be = backend or make_backend(out, cfg, spec, solver_name)
    limit = be.largest_problem()
    log(f"[anneal] backend={be.spec} solver={be.solver_id()} largest_problem={limit}")
    units = list(anneal_units(out, cfg, keys, Ns=Ns))
    too_big = sorted({u[3] for u in units if limit is not None and u[3] > limit})
    if too_big:
        log(f"[anneal] skipping N={too_big}: above the largest embeddable clique ({limit})")
    todo = [u for u in units if (limit is None or u[3] <= limit)
            and not anneal_path(out, be.label, u[0], u[3], u[4]).exists()]
    log(f"[anneal] {len(units)} calls in the campaign, {len(todo)} still to do")
    if check_only:
        _estimate_qpu_time(be, todo, log)
        return
    total_access = 0.0
    budget = budget or QPUBudget(None)
    for n, (key, eps, J0, N, b) in enumerate(todo):
        if max_calls is not None and n >= max_calls:
            log(f"[anneal] stopped after --max-calls {max_calls}; rerun to continue")
            break
        est = _call_estimate(be, N, cfg)
        if not budget.allows(est):
            log(f"[anneal] budget: spent {budget.spent:.3f}s of {budget.limit:.3f}s; next call (N={N}) estimated
 "
                f"{est:.3f}s x safety {budget.factor():.2f} would exceed it -> stopped (rerun to continue)")
            break
        J = ensure_instances(out, cfg, key, eps, J0, N)
        P = PlannerProblem(J[b])
        if cfg.chain_strength_by_N:                  # per-N prefactor chosen by coupling-scan
            from coupling_scan import coupling_for
            be.settings.chain_strength_prefactor = coupling_for(cfg.chain_strength_by_N, be.label, N,
                                                                cfg.chain_strength_prefactor)
        for attempt in range(retries + 1):
            try:
                res = be.sample(P, rng=np.random.SeedSequence([cfg.seed, 3, N, b]),
                                label=f"skgame {out.name} {be.label} {key} N={N} i={b}")
                break
            except Exception as exc:                     # network hiccups, solver busy, ...
                if attempt == retries or not _transient(exc):
                    raise
                log(f"[anneal] {key} N={N} i={b}: {type(exc).__name__}: {exc}; retry in {2 ** (attempt + 1)}s")
                time.sleep(2 ** (attempt + 1))
        res.update(key=key, eps=eps, J0=J0, N=N, instance=b, backend=be.spec)
        res["info"]["estimated_access_s"] = est
        phys = res.pop("_physical", None)
        if phys is not None:                         # the exact physical problem sent (before auto_scale)
            pp = anneal_path(out, be.label, key, N, b).with_name(f"N{N:04d}_i{b}_physical.npz")
            pp.parent.mkdir(parents=True, exist_ok=True)
            np.savez_compressed(pp, **phys)
            res["info"]["physical_problem_file"] = pp.name
        _write_json(anneal_path(out, be.label, key, N, b), res)
        acc = res["info"].get("timing_us", {}).get("qpu_access_time", 0.0) / 1e6
        budget.charge(acc if acc > 0 else est, est)
        total_access += acc
        cbf = res["info"].get("chain_break_fraction_mean")
        log(f"[anneal] {n + 1:4d}/{len(todo)} {be.label} {key:10s} N={N:4d} i={b} best={res['best_reward']:.5f}
 "
            f"chain_breaks={'-' if cbf is None else f'{cbf:.3f}'} qpu_access={acc * 1e3:.1f}ms (est {est * 1e3
:.0f}ms) "
            f"(total {total_access:.3f}s)")
```

Rule: before each call, with $\hat{t}$ the level-2 estimate,

$$\text{call allowed} \iff t_{\text{spent}} + f\,\hat{t} \le S, \qquad f = \max\left(1.5,\ 1.1 \cdot \max_{\text{past calls}} t_{\text{actual}}/\hat{t}\right). \tag{4}$$

- The guard starts with a 50% safety margin. If the real charges turn out larger than the model, the margin grows automatically.

- After every call t_{spent} grows by the *real* `qpu_access_time`.
- The budget covers all back-ends of one command (Zephyr, then Pegasus).
- When a call is refused, the command stops cleanly. Everything done so far is saved, and the same command continues later (resume).
- Without a timing model (mock QPU) the estimate is a deliberately pessimistic formula: $20\,\text{ms} + n_{\text{reads}}(t_a + 300\,\mu\text{s})$.

It is tested in `tests/test_campaign.py`: the rule itself, a 0.01 s budget that refuses every call, and a 1 s budget that completes and records estimates.

What the guard cannot guarantee. It caps the time *this code* spends, using D-Wave's own numbers. It cannot stop a call once submitted, and it does not see time spent by other programs or other users of the same Leap project. The Leap dashboard remains the authority on what is left.

8 Broken chains, read by read

What is saved. For every call, `sampleset_statistics` (listing above) stores:

- `reads`: one row per distinct pair (configuration, number of broken chains), with the reward, the reward after greedy polishing, the number of broken chains and the number of reads;
- `info.broken_chains_hist`: number of reads with $0, 1, 2, \ldots$ broken chains;
- `info.chain_break_count`: for each logical spin i, in how many reads its chain broke. With equal-length clique chains this should be roughly uniform; outliers point at faulty qubits or couplers;
- `info.chain_lengths`, `info.chain_strength`, `info.discarded_reads` (only with `chain_break_method=discard`).

A chain of spin i is *broken* in a read when its qubits do not all agree. Its logical value is then decided by the chain-break method (majority vote by default). The fraction D-Wave reports, `chain_break_fraction`, is exactly (broken chains)$/N$, and the code checks this.

Analysis. `python main_simulation.py chains --out DIR` (also run by `analyze`, and usable before the classical stage) writes `summary/chain_breaks.csv` and `chain_breaks.png`. For each back-end, N and number of broken chains k they give the fraction of reads, the success probability, and the mean residual, raw and after polishing. The reference is the exact or classical optimum when available, otherwise the best read seen.

`main_simulation.py`, lines 726–781: solution quality vs number of broken chains

```
726 def chain_break_table(out: Path, cfg: CampaignConfig) -> list[dict]:
727     """Solution quality as a function of the number of broken chains in the read.
728 
729     One row per (back-end, problem class, N, number of broken chains): how many reads, and among
730     them the success probability and the mean residual, raw and after greedy polishing. The
731     reference is the same as in `collect` (exact / classical, improved by any annealer read);
732     if the classical stage has not run yet it is the best read of all annealers ("best seen").
733     """
734     if not (out / "anneal").exists():
735         return []
736     labels = sorted(p.name for p in (out / "anneal").glob("*"))
737     rows = []
738     for key, eps, J0 in _classes(cfg):
739         for N in cfg.Ns:
740             recs = {lab: [json.loads(p.read_text()) if p.exists() else None
741                           for p in (anneal_path(out, lab, key, N, b) for b in range(cfg.n_instances))]
742                     for lab in labels}
```

```
            if not any(r for v in recs.values() for r in v):
                continue
            cp = classical_path(out, key, N)
            if cp.exists():
                c = json.loads(cp.read_text())
                ref, src = np.array(c["ref"], float), ("exact" if c["ref_exact"] else "classical")
            else:
                ref, src = np.full(cfg.n_instances, -INF), "best_seen"
            for v in recs.values():
                for b, r in enumerate(v):
                    if r is not None:
                        ref[b] = max(ref[b], r["best_reward"], r.get("polished_best_reward", -INF))
            for lab, v in recs.items():
                acc = {}
                for b, r in enumerate(v):
                    if r is None or "reads" not in r:
                        continue
                    t = r["reads"]
                    rw, nb, ct = (np.array(t[k], float) for k in ("reward", "n_broken", "count"))
                    pr = np.array(t["polished_reward"], float) if t.get("polished_reward") else None
                    ok = _success(rw, ref[b])
                    okp = _success(pr, ref[b]) if pr is not None else None
                    for k in np.unique(nb).astype(int):
                        m = nb == k
                        a = acc.setdefault(k, {"reads": 0, "succ": 0.0, "succ_p": 0.0, "res": 0.0, "res_p":
    0.0})
                        a["reads"] += ct[m].sum()
                        a["succ"] += ct[m][ok[m]].sum()
                        a["res"] += np.sum(ct[m] * (ref[b] - rw[m]) / abs(ref[b]))
                        if pr is not None:
                            a["succ_p"] += ct[m][okp[m]].sum()
                            a["res_p"] += np.sum(ct[m] * (ref[b] - pr[m]) / abs(ref[b]))
                tot = sum(a["reads"] for a in acc.values())
                for k, a in sorted(acc.items()):
                    rows.append({"backend": lab, "problem": key, "N": N, "n_broken": k, "reads": int(a["reads"])
    ,
                                 "frac_reads": a["reads"] / tot, "p_success": a["succ"] / a["reads"],
                                 "p_success_polished": a["succ_p"] / a["reads"],
                                 "residual_mean": a["res"] / a["reads"], "residual_polished_mean": a["res_p"] /
    a["reads"],
                                 "reference": src})
        return rows
```

Example (classical simulator, deliberately weak chains $p = 0.5$). At $N = 16$ on Pegasus, 70% of the reads have no broken chain and hit the optimum in 88% of cases. Reads with one or more broken chains never do, raw. Greedy polishing brings those with 1–2 broken chains back to 84–96%. The residual grows almost linearly with the number of broken chains. On the real QPU, with the default $p = 1.414$, the test of 27 October had chain-break fractions of 0 and 0.001 at $N \leq 16$.

9 Ground states

`python main_simulation.py ground --out DIR` (also run by `analyze`) writes `summary/ground_states.csv`. For every instance it gives:

- the best configuration found (a string of $+/-$, one per agent), with the sign fixed so that agent 0 is $+$, since S and $-S$ have the same energy;
- its reward R_N and its energy $H = -\frac{N}{2}(R_N - \text{offset})$;
- where it came from, and whether it is **certified** (exact enumeration, $N \leq 24$) or only **best known**;
- for every source (classical reference, each annealer raw and polished) whether that source found it, and its Hamming distance from it.

The classical stage now saves the reference *configuration* (`ref_S`, `ref_source`) and not only its value. Each QPU result file already holds `best_S` and `polished_best_S`.

main_simulation.py, lines 853–915: the ground-state report

```
def ground_states(out: Path, cfg: CampaignConfig, log=print) -> list[dict]:
    """The ground state (best configuration) of every instance, from every source available.

    Candidates: the classical reference (exact Gray-code enumeration for N <= n_exact: certified;
    otherwise the best classical solver / long parallel tempering: best known) and, for every
    annealing back-end, its best read and its best polished read. The best candidate is reported
    with its reward R_N, energy H = -N (R_N - offset)/2, configuration (+/- per agent, sign fixed
    so that agent 0 is +, since S and -S are equivalent), where it came from, whether it is
    certified, and for each source whether it found it and how far (Hamming distance) its best
    configuration is. Writes summary/ground_states.csv."""
    labels = sorted(p.name for p in (out / "anneal").glob("*")) if (out / "anneal").exists() else []
    rows = []
    for key, eps, J0 in _classes(cfg):
        for N in cfg.Ns:
            cp = classical_path(out, key, N)
            c = json.loads(cp.read_text()) if cp.exists() else None
            if c is None and not any(anneal_path(out, l, key, N, 0).exists() for l in labels):
                continue
            J = ensure_instances(out, cfg, key, eps, J0, N)
            for b in range(cfg.n_instances):
                P = PlannerProblem(J[b])
                cand = []                                   # (reward, spins, column, source)
                if c is not None and c.get("ref_S"):
                    cand.append((c["ref"][b], c["ref_S"][b], "classical", c["ref_source"][b]))
                for lab in labels:
                    ap = anneal_path(out, lab, key, N, b)
                    if ap.exists():
                        r = json.loads(ap.read_text())
                        if r.get("best_S"):
                            cand.append((r["best_reward"], r["best_S"], lab, lab))
                        if r.get("polished_best_S"):
                            cand.append((r["polished_best_reward"], r["polished_best_S"], lab + "+polish", lab +
  "+polish"))
                if not cand:
                    continue
                top = max(t[0] for t in cand)               # ties (to rounding) go to the first source:
 classical/exact
                best = next(t for t in cand if t[0] >= top - TOL * max(1, abs(top)))
                g = _canonical(best[1])
                S = np.array([1.0 if ch == "+" else -1.0 for ch in g])
                certified = bool(c is not None and c.get("ref_exact") and c.get("ref_S"))
                row = {"problem": key, "N": N, "instance": b, "reward": float(P.reward(S)),
                       "energy": float(P.energy(S)), "ground_state": g, "found_by": best[3],
                       "certified": certified, "status": "exact optimum" if certified else "best known"}
                for rw, sp, col, _src in cand:
                    row[f"{col}:found"] = bool(rw >= top - TOL * max(1, abs(top)))
                    row[f"{col}:hamming"] = sum(x != y for x, y in zip(_canonical(sp), g))
                rows.append(row)
    if rows:
        path = out / "summary" / "ground_states.csv"
        path.parent.mkdir(parents=True, exist_ok=True)
        cols = []
        for r in rows:
            cols += [k for k in r if k not in cols]
        with open(path, "w", newline="") as f:
            w = csv.DictWriter(f, fieldnames=cols)
            w.writeheader()
            w.writerows(rows)
        for r in rows:
            log(f"[ground] {r['problem']:10s} N={r['N']:4d} i={r['instance']} R_N={r['reward']:.5f} "
                f"H={r['energy']:9.4f} {r['status']:13s} by {r['found_by']:24s} {r['ground_state'] if r['N'] <=
 40 else r['ground_state'][:40] + '...'}")
        log(f"[ground] -> {path}")
    else:
        log("[ground] no results yet")
    return rows
```

When the QPU's graph changes. A QPU's working graph changes over time as qubits and couplers go offline or come back. On 2 November the Zephyr chip's largest clique was 105, against 104 on 27 October. An embedding cached on an earlier day may then contain a chain that is no longer connected (`DisconnectedChainError`). `EmbeddingCache.get` therefore re-validates every

cached embedding against the current graph (`is_valid_embedding`), recomputes it if it no longer fits, and keeps the old file as `N....stale.json`. Errors in the embedding or the problem are no longer retried (`_transient`), and they occur before any QPU time is used.

10 Chain strength and coupling scale: what is set, and what is not known

What is set. The chain strength is $c = p\, J_{\mathrm{rms}}\sqrt{N-1} \approx p$ for SK, with the prefactor p (`chain_strength_prefactor`, default 1.414, overridden per N by `coupling-scan`) the only free knob. Why p must be larger for the all-to-all problem: `prefactor_and_shimming.pdf`. The logical couplings W_{ij} are split evenly over the $|E_{ij}|$ couplers between chains (about 1.1–1.2), and `auto_scale` divides the whole physical problem by its largest $|J|$. With the default p that is the chain coupling c, so the SK couplings end up at about 0.05 on $[-1, 1]$. With a small p (e.g. 0.5) the largest coupling can be a logical one, as the tests show, and then the logical couplings are no longer the ones compressed by c.

What is not known in advance. There is no formula for the best p:

- too small, and chains break: their spins are guessed by the chain-break method;
- too large, and chains are rigid and the SK part is squeezed towards the analogue noise.

The optimum depends on N, on the instance class and on the chip. The same holds for the chain-break method and the annealing time. The default is a well-founded starting point, not an optimum. On the QPU it can only be found empirically: run a few values of p on the same instances and compare the success rate together with the broken-chain statistics. `scan-summary` does this comparison across campaign directories that differ only in p (README_LOCAL.md gives the commands).

What is recorded to judge it. For every call:

- c and the auto-scale divisor;
- the per-read broken chains and which chains break;
- the exact physical problem sent (couplers and values, `*_physical.npz`).

A rule of thumb from D-Wave's guidance: chain-break fractions of a few per cent or less, with the success rate still rising as p decreases, mean there is room to lower p.

11 Choosing the chain strength for every N: `coupling-scan`

For every QPU and every N of a (reduced) grid, the same instance is annealed with each prefactor p of a grid, by default $0.75, 1, 1.414, 2, 3, 4, 6$, with 200 reads. The grid reaches 6 because the chains of the all-to-all SK problem need $p \approx 3$–4 at $N \gtrsim 30$: the derivation, the upper bound set by the control error, the extended coupler range and the flux-bias shimming (`shim`) are in `prefactor_and_shimming.pdf`. The scan counts the broken chains of every read.

The rule. Take the *smallest* p whose mean fraction of broken chains is at most a threshold (default 0: no chain broken in any read). If no p reaches it, take the p with the fewest broken chains. Minimising the breaks alone would always select the largest p tested, since breaks decrease monotonically with p, while a larger p compresses the SK couplings further (auto-scale divides by $\approx c$). The smallest p that keeps the chains intact is the standard compromise. The output flags a choice at the edge of the grid.

coupling_scan.py, lines 36–49: the choice of p

```python
def choose_coupling(rows: list[dict], threshold: float = 0.0) -> tuple[float, str]:
    """rows: [{"p", "broken_fraction"}...] for ONE back-end and N. Returns (p, reason)."""
    ok = [r for r in rows if r["broken_fraction"] <= threshold + 1e-15]
    if ok:
        best = min(ok, key=lambda r: r["p"])
        why = f"smallest p with broken-chain fraction <= {threshold:g}"
        if best["p"] == min(r["p"] for r in rows):
            why += " (already at the SMALLEST p tested: a smaller p may also work)"
        return best["p"], why
    best = min(rows, key=lambda r: (r["broken_fraction"], r["p"]))
    why = "no p reached the threshold: fewest broken chains"
    if best["p"] == max(r["p"] for r in rows):
        why += " (at the LARGEST p tested: extend the grid upwards)"
    return best["p"], why
```

Use in the main run. The table $\{\text{QPU} : \{N : p\}\}$ is stored in `config.json` (`chain_strength_by_N`). The QPU stage sets p call by call: exact for scanned N, linear interpolation in between, the last value beyond the largest scanned N. Every result records the p actually used, and `validate` checks the chain strength of each call against its own p.

coupling_scan.py, lines 52–60: prefactor for a given N

```python
def coupling_for(table: dict, label: str, N: int, default: float) -> float:
    """Prefactor for back-end 'label' at size N from a {label: {N: p}} table: exact N, else linear
    interpolation between the nearest scanned N, else the nearest end (constant extrapolation)."""
    t = table.get(label) or {}
    if not t:
        return default
    xs = sorted(int(k) for k in t)
    ys = [float(t[str(x)] if str(x) in t else t[x]) for x in xs]
    return float(np.interp(N, xs, ys))
```

Cost. The scan costs ≈ 5.4 s of QPU time for $N = 8, \ldots, 48$ and 104 on both QPUs. That is about 15.8 ms of programming plus 200 reads per call, for 98 calls. The time does not depend on p.

12 Reproducibility: everything needed to reconstruct a call

Each call saves:

- the problem id and label (findable in the Leap dashboard);
- the solver identity, ranges and a fingerprint of the working graph at that moment;
- the embedding actually used;
- the chain strength;
- the exact physical problem (`N...._physical.npz`: qubits, couplers, values before scaling) and the auto-scale divisor;
- every distinct configuration read, with its reward, polished reward and number of broken chains.

The instances (full J matrices) and the configuration are saved once per campaign. `RESULTS_FORMAT.md` documents every field.

`python main_simulation.py validate --out DIR` re-derives everything from these inputs:

- the reward of every stored configuration, and the exact optimum for $N \leq 24$;
- the read counts, and the broken-chain counts against D-Wave's `chain_break_fraction`;
- the chain strength from the formula;

- the physical problem: re-embedding the instance with the saved embedding and chain strength reproduces the saved one exactly;
- that every embedding is a valid minor of K_N on the solver's own topology.

main_simulation.py, lines 918–1025: validate

```python
def validate(out: Path, cfg: CampaignConfig, log=print) -> dict:
    """Re-derive every stored number from the stored inputs, a posteriori.

    For each annealer result: rebuild the instance from instances/*.npz and recompute the reward of
    every stored configuration (reads.S, best_S, polished_best_S); check read counts, the broken-
    chain histogram against D-Wave's chain_break_fraction, the chain strength against
    p * J_rms * sqrt(N-1), the embedding_used (valid minor of K_N on the ideal graph of the
    topology, which contains every working graph) and its chain lengths. For each classical
    file: recompute the reward of the reference configuration and, for N <= n_exact, re-run the
    exact enumeration. Returns {check: [passed, failed]}."""
    from skgame.optim.embedding import is_valid_embedding, topology as topo
    from skgame.optim.backends import parse_spec
    tally = {}

    def check(name, ok, detail=""):
        t = tally.setdefault(name, [0, 0])
        t[0 if ok else 1] += 1
        if not ok:
            log(f"[validate] FAILED {name}: {detail}")

    def close(a, b):
        return abs(a - b) <= 1e-9 * max(1.0, abs(b))

    graphs = {}
    for key, eps, J0 in _classes(cfg):
        for N in cfg.Ns:
            if not instance_path(out, key, N).exists():
                continue
            J = ensure_instances(out, cfg, key, eps, J0, N)
            cp = classical_path(out, key, N)
            if cp.exists():
                c = json.loads(cp.read_text())
                for b in range(cfg.n_instances):
                    P = PlannerProblem(J[b])
                    if c.get("ref_S"):
                        S = np.array([1.0 if ch == "+" else -1.0 for ch in c["ref_S"][b]])
                        check("classical reference configuration -> reference value", close(P.reward(S), c["ref"
    ][b]),
                              f"{key} N={N} i={b}")
                    if c["ref_exact"] and N <= 24:
                        check("exact optimum re-computed", close(ExactEnumeration(max_N=24).solve(P).reward, c["
    ref"][b]),
                              f"{key} N={N} i={b}")
            for lab in sorted(p.name for p in (out / "anneal").glob("*")) if (out / "anneal").exists() else []:
                for b in range(cfg.n_instances):
                    ap = anneal_path(out, lab, key, N, b)
                    if not ap.exists():
                        continue
                    r = json.loads(ap.read_text())
                    info, P, where = r["info"], PlannerProblem(J[b]), f"{lab} {key} N={N} i={b}"
                    to_S = lambda st: np.array([1.0 if ch == "+" else -1.0 for ch in st])
                    if r.get("best_S"):
                        check("best configuration -> best reward", close(P.reward(to_S(r["best_S"])), r["
    best_reward"]), where)
                    if r.get("polished_best_S"):
                        check("polished configuration -> polished reward",
                              close(P.reward(to_S(r["polished_best_S"])), r["polished_best_reward"]), where)
                    t = r.get("reads")
                    if t and t.get("S"):
                        rec = np.atleast_1d(P.reward(np.array([to_S(x) for x in t["S"]]))) if t["S"] else np.
    zeros(0)
                        check("every stored configuration -> its reward", np.allclose(rec, t["reward"], rtol=1e
    -9, atol=1e-12), where)
                    if t:
                        check("read counts add up", sum(t["count"]) == info["num_reads"] - info.get("
    discarded_reads", 0)
                              or sum(t["count"]) == info["num_reads"], where)
                    h = info.get("broken_chains_hist")
```

```
                    if h and "chain_break_fraction_mean" in info and not info.get("discarded_reads"):
                        mean = sum(int(k) * v for k, v in h.items()) / max(1, sum(h.values())) / N
                        check("broken-chain counts agree with D-Wave's chain_break_fraction",
                              abs(mean - info["chain_break_fraction_mean"]) < 1e-9, where)
                    if info.get("chain_strength") is not None:
                        q = P.W[np.triu_indices(N, 1)]
                        expect = info["settings"]["chain_strength_prefactor"] * np.sqrt(np.mean(q ** 2)) * np.
    sqrt(N - 1)
                        check("chain strength = p J_rms sqrt(N-1)", close(info["chain_strength"], expect), where
    )
                    pf = ap.with_name(info["physical_problem_file"]) if info.get("physical_problem_file") else
    None
                    if pf is not None and pf.exists() and info.get("embedding_used"):
                        import dimod
                        from dwave.embedding import EmbeddedStructure
                        ph = np.load(pf)
                        emb = {int(k): v for k, v in info["embedding_used"].items()}
                        edges = list(zip(ph["u"].tolist(), ph["v"].tolist()))
                        rebuilt = EmbeddedStructure(edges, emb).embed_bqm(P.to_bqm(), chain_strength=info["
    chain_strength"],
                                                                          smear_vartype=dimod.SPIN)
                        saved = dimod.BinaryQuadraticModel(dict(zip(ph["h_q"].tolist(), ph["h"].tolist())),
                                                           dict(zip(edges, ph["J"].tolist())), float(ph["offset"
    ]), "SPIN")
                        check("physical problem = embed(instance, embedding_used, chain_strength)",
                              rebuilt.is_almost_equal(saved), where)
                        sc = info["programmed"].get("scaling", {"mode": "auto_scale"})
                        if sc["mode"] == "extended_j_range":
                            own = {q: i for i, ch in emb.items() for q in ch}
                            chain = np.array([abs(v) for (x, y), v in zip(edges, ph["J"]) if own[x] == own[y]])
                            logi = np.array([abs(v) for (x, y), v in zip(edges, ph["J"]) if own[x] != own[y]])
                            expect = max(chain.max() / -sc["ext_lo"], logi.max() if logi.size else 0.0)
                            check("extended-range divisor = max(c/|ext_lo|, max logical |J|)",
                                  close(sc["divisor"], expect), where)
                        else:
                            check("auto_scale divisor = max |J| of the physical problem",
                                  close(float(np.abs(ph["J"]).max()), info["programmed"]["
    max_abs_J_before_scaling"]), where)
                    if info.get("embedding_used"):
                        emb = {int(k): v for k, v in info["embedding_used"].items()}
                        t = (info.get("solver_properties") or {}).get("topology") or {}
                        tp = t.get("type") or parse_spec(info["backend"])[1]
                        shape = tuple(t["shape"]) if t.get("shape") else topo(tp).shape
                        from skgame.optim.embedding import Topology
                        G = graphs.setdefault((tp, shape), Topology(tp, shape, "").ideal_graph())
                        check("embedding is a valid minor of K_N", is_valid_embedding(emb, N, G), where)
                        check("chain lengths match the embedding", [len(emb[i]) for i in range(N)] == info["
    chain_lengths"], where)
                for name, (ok, bad) in tally.items():
                    log(f"[validate] {'OK  ' if not bad else 'FAIL'} {name}: {ok} passed, {bad} failed")
                if not tally:
                    log("[validate] nothing to validate yet")
                return tally
```

13 Ising or QUBO? And is the problem trivial?

Ising, which is equivalent to QUBO. The QPU's native form is Ising: spins $S_i = \pm 1$, fields h_i, couplings J_{ij}. The project sends the SK problem in this form, with $h = 0$ and $J_{ij} = -W_{ij}$ on all $N(N-1)/2$ pairs (`to_bqm`). A QUBO is the same model in binary variables $x_i \in \{0, 1\}$. With $S_i = 2x_i - 1$:

$$\begin{aligned} H &= -\sum_{i<j} W_{ij} S_i S_j = -\sum_{i<j} W_{ij}(2x_i - 1)(2x_j - 1) \\ &= \sum_{i<j} (-4W_{ij})\, x_i x_j + \sum_i \Big(2\sum_{j\neq i} W_{ij}\Big) x_i - \sum_{i<j} W_{ij}. \end{aligned} \tag{5}$$

So $Q_{ij} = -4W_{ij}$, $Q_{ii} = 2\sum_{j\neq i} W_{ij}$, and the offset is $-\sum_{i<j} W_{ij}$. The minimiser is the same (via $x = (S+1)/2$), and so is every energy. `PlannerProblem.to_qubo` returns this form. A test

checks it against the Ising form on all 2^9 configurations of an instance, and checks that dimod's `change_vartype` turns one into the other. Sending the QUBO instead would change nothing: dimod converts it to the same Ising problem.

`skgame/optim/problem.py`, lines 52–64: the QUBO form

```
52      def to_qubo(self):
53          """The same problem as a QUBO in binary variables x_i in {0, 1}, with S_i = 2 x_i - 1.
54
55          H(S) = -sum_{i<j} W_ij S_i S_j
56               = -sum_{i<j} W_ij (2x_i - 1)(2x_j - 1)
57               = sum_{i<j} (-4 W_ij) x_i x_j + sum_i (2 sum_{j != i} W_ij) x_i - sum_{i<j} W_ij,
58          so Q_ij = -4 W_ij (i < j), Q_ii = 2 sum_{j != i} W_ij, offset = -sum_{i<j} W_ij.
59          Same minimiser (mapped by x = (S + 1) / 2) and same energy. The QPU works in Ising form;
60          dimod converts between the two exactly (bqm.change_vartype)."""
61          iu = np.triu_indices(self.N, 1)
62          Q = {(int(i), int(j)): -4.0 * float(self.W[i, j]) for i, j in zip(*iu)}
63          Q.update({(i, i): 2.0 * float(self.W[i].sum()) for i in range(self.N)})
64          return Q, -float(self.W[iu].sum())
```

Is it trivial? The problem is not:

- the SK ground state is a dense spin glass with frustration and exponentially many local minima. Its worst case is NP-hard, and typical instances need exponential time for every known exact method;
- there is no field ($h = 0$), so S and $-S$ are both ground states. The code counts either as success.

What *is* trivial is the 1-second test: $N = 8, 16$ is solved exactly by a laptop in milliseconds. It tests the machinery (token, embedding, timing, budget), not the QPU's ability. The campaign's real question starts at large N, up to 104 on Zephyr and 177 on Pegasus. There, good classical solvers (parallel tempering, tabu) remain very strong on dense SK, and the QPU is handicapped by the all-to-all embedding: long chains, couplings compressed to ~ 0.05, analogue noise. The honest expectation is not that the QPU wins. The campaign measures where it stops finding the optimum, how that depends on broken chains, and whether Zephyr's shorter chains move that point. The `ferro_easy` regime ($J_0 = 3$) is *deliberately* easy: it is the control case where every method should succeed.

14 From reads to results

`main_simulation.py`, lines 519–550: per back-end metrics in `collect`

```
519             for s, recs in ann.items():
520                 done = [(b, r) for b, r in enumerate(recs) if r is not None]
521                 if not done:
522                     continue
523                 p = np.array([np.sum(np.array(r["counts"])[_success(r["rewards"], ref[b])]) / max(1, np.sum(r["
    counts"]))
524                               for b, r in done])
525                 best_res = np.array([(ref[b] - r["best_reward"]) / abs(ref[b]) for b, r in done])
526                 mean_res = np.array([np.average((ref[b] - np.array(r["rewards"])) / abs(ref[b]), weights=r["
    counts"])
527                                      if np.sum(r["counts"]) else np.nan for b, r in done])
528                 t_read = np.median([_time_per_read(r) for b, r in done])
529                 acc = [r["info"].get("timing_us", {}).get("qpu_access_time", np.nan) / 1e6 for b, r in done]
530                 cbf = [r["info"].get("chain_break_fraction_mean", np.nan) for b, r in done]
531                 extra = {}
532                 if all("polished_rewards" in r for b, r in done):
533                     pp = np.array([np.sum(np.array(r["polished_counts"])[_success(r["polished_rewards"], ref[b])
    ])
534                                    / max(1, np.sum(r["polished_counts"])) for b, r in done])
535                     extra.update(p_success_polished=float(np.median(pp)),
536                                  residual_polished=float(np.median([(ref[b] - r["polished_best_reward"]) / abs(
    ref[b])
```

```
                                                                    for b, r in done])))
                emb = [r["info"].get("embedding") for b, r in done]
                if all(emb):
                    extra.update(chain_len_mean=float(np.mean([e["chain_len_mean"] for e in emb])),
                                 chain_len_max=int(max(e["chain_len_max"] for e in emb)),
                                 n_qubits=int(np.mean([e["n_qubits"] for e in emb])),
                                 chain_strength=_nan_to_none(float(np.mean([r["info"].get("chain_strength") or
    np.nan
                                                                            for b, r in done]))))
                rows.append(dict(base, kind="anneal", method=f"anneal_{s}", regime="", n_done=len(done), **extra
    ,
                                 p_success_median=float(np.median(p)), p_success_mean=float(p.mean()),
                                 p_best_found=float(np.mean(best_res <= TOL)),
                                 residual_median=float(np.median(best_res)), residual_mean_read=float(np.
    nanmedian(mean_res)) if np.isfinite(mean_res).any() else None,
                                 seconds_median=t_read, tts99=_tts(p.mean(), t_read),
                                 qpu_access_s=float(np.nansum(acc)), chain_break_fraction=_nan_to_none(float(np.
    mean(cbf)))))
```

For each problem class, N and instance b, with reference R_{ref} (exact for $N \le 24$; otherwise the best of the classical solvers and of every annealer read, polished reads included):

$$p_{\text{success}} = \frac{\#\{\text{reads with } R \ge R_{\text{ref}} - 10^{-9}\}}{n_{\text{reads}}} \quad \text{(per read)}, \tag{6}$$

$$\text{residual} = \frac{R_{\text{ref}} - R_{\text{best read}}}{|R_{\text{ref}}|}, \tag{7}$$

$$t_{\text{read}} = \frac{t_{\text{sampling}}}{n_{\text{reads}}} \quad (t_{\text{sampling}} = \texttt{qpu_sampling_time}), \tag{8}$$

$$\text{TTS}_{99} = t_{\text{read}} \frac{\ln 0.01}{\ln(1-p)}. \tag{9}$$

`main_simulation.py`, lines 569–584: time per read and TTS_{99}

```
def _time_per_read(r) -> float:
    """QPU: sampling time per read (anneal + readout + delay); simulator: wall time per read."""
    t = r["info"].get("timing_us", {})
    n = r["info"]["num_reads"]
    if "qpu_sampling_time" in t:
        return t["qpu_sampling_time"] / 1e6 / n
    return r["info"]["wall_seconds"] / n


def _tts(p: float, t: float) -> float:
    """Time to reach the reference with 99 % confidence: t ln(0.01) / ln(1 - p)."""
    if p >= 1:
        return float(t)
    if p <= 0:
        return INF
    return float(t * math.log(0.01) / math.log1p(-p))
```

`compare_backends` ranks the back-ends of the same device for each (class, N): higher median p_{success} wins, ties go to the lower residual. The output is `summary/backend_comparison.csv` and `.png`.

15 How the D-Wave path was tested

- `test_embedding.py`: clique embeddings are valid on both topologies (every chain connected, chains disjoint, every pair linked); Zephyr chains are shorter at production size; the cache round-trips; back-end specs parse; the embedded simulator finds the exact ground state, and the chain strength equals $p\,J_{\text{rms}}\sqrt{N-1}$ computed independently.
- `test_campaign.py`: every read is kept and `seed` is skipped where unsupported (with `ExactSolver`); the whole campaign runs end to end and resumes; **both QPU back-ends run against `MockDWaveSampler`** with Zephyr and Pegasus topologies, using the real `DWaveSampler` code path from `solver={topology__type}` onwards; the budget guard (rule, refusal, completion, `--spent`) is checked.

- What the mock cannot test: the real timing values, the real noise, SAPI errors and Leap accounting. That is what the one-second test below is for.

16 The one-second test, and a plan for the 180 s

16.1 Test: at most 1 s of QPU time, both topologies

A tiny separate campaign: $N = 8$ and 16, one instance, one problem class, 100 reads per call, both QPUs. That makes 4 calls. At 100 reads each the timing model gives a few tens of ms per call, so the whole test costs a fraction of a second. The guard makes **1 s a hard ceiling** anyway.

```
cd ...\Dimod\src\unlearnable_games_explanation      # (PowerShell, .venv active)
$T = "results\qpu_test"
python main_simulation.py init --preset smoke --out $T --Ns 8,16 --n-instances 1 --regimes
    sg_satisficing --num-reads 100
python main_simulation.py qpu --out $T --check                  # predicted cost: spends
    nothing
python main_simulation.py qpu --out $T --budget-seconds 1       # the test: <= 1 s,
    normally ~0.1-0.2 s
python main_simulation.py qpu --out $T --spent                  # what Leap charged, and
    actual/estimate
python main_simulation.py classical --out $T                    # seconds, for the
    reference optimum
python main_simulation.py analyze --out $T                      # success per read, chain
    breaks
```

What to check in the output:

1. every call prints `qpu_access=...ms (est ...ms)`: real vs predicted;
2. `--spent`: the total, well below 1 s, and `actual/estimate` close to 1;
3. `chain_breaks`: should be 0 or tiny at $N = 8, 16$;
4. the Leap dashboard shows the same usage.

16.2 Spending the 180 s before 4 November

Your check gives 157.3 s for the full campaign on both topologies at 1000 reads. That leaves about 22 s of margin, *if* the real charges match the model, which the test will show. Two safe options:

plan	QPU time	comment
`full`, 1000 reads, both QPUs	≈ 157 s	as designed; run with `--budget-seconds 170` so it can never pass 170 s
`full`, 500 reads (`init ... --num-reads 500`)	≈ 80 s	time $\propto$ reads; leaves ≈ 90 s for a chain-strength scan at the N where success drops

```
$F = "results\full"
python main_simulation.py init --preset full --out $F --force          # (add --num-reads
    500 for option 2)
python main_simulation.py qpu --out $F --check
python main_simulation.py qpu --out $F --budget-seconds 170            # stops by itself
    before 170 s
python main_simulation.py qpu --out $F --spent
```

The QPU stage does not need the classical stage. That can run before or after, on the laptop or on Leonardo, and `analyze` combines them. If the run stops on the budget, it has saved everything done so far, and `--spent` shows the total.

References

- D-Wave, Ocean documentation: https://docs.dwavequantum.com/en/latest/ocean/; Leap authorisation: https://docs.dwavequantum.com/en/latest/ocean/leap_authorization.html.
- D-Wave, QPU timing and `estimate_qpu_access_time`: `dwave.cloud.solver.StructuredSolver` (source read for §6, dwave-cloud-client 0.14.8).
- Boothby, King, Raymond, *Fast clique minor generation in Chimera qubit connectivity graphs*, Quantum Inf. Process. 15, 495 (2016): the clique embedding.
- Cai, Macready, Roy, *A practical heuristic for finding graph minors*, arXiv:1406.2741 (2014): minorminer.
- Cineca, Quantum Computing as a Service: https://www.quantumcomputinglab.cineca.it/en/resources.
- Project: `Dimod/explanation_project.pdf` (SK game, algorithms, embedding, chains, noise), `Dimod/LEAP_TOKEN.md`, `README_LOCAL.md`, `README_CINECA.md`.